

ME 571 Final Project Report: Smart Biopsy Training Device with Real-Time Feedback and Active Guidance

Group 7: Noelle Flanagan, Isara Cholaseuk, Tanish Katial, Shresth Verma, Sauman Raaj Aarthi Ganesh Vetrivel

A. Significance. Minimally invasive procedures such as needle biopsies are essential diagnostic tools for conditions ranging from cancer to chronic liver disease [1]. Accuracy and safety of these procedures are highly dependent on precise needle placement [2]. Inaccurate biopsies can lead to serious clinical consequences, including missed diagnoses, hemorrhage, pneumothorax, or damage to surrounding structures [3]. These risks are especially pronounced when procedures are performed by inexperienced clinicians who have underdeveloped skills [4]. Improving proficiency requires extensive hands-on practice to master delicate motor control, proper insertion angle, and tactile feedback skills that are difficult to acquire using current static training models or cadaver-based simulations [5]. These methods are often limited by high cost, lack of real-time feedback, and absence of corrective guidance, resulting in prolonged training time and increased risk of patient harm during early clinical practice.

Our proposed smart biopsy simulation device addresses a critical gap in procedural training by offering an interactive platform that delivers real-time feedback on user performance. By integrating pressure and orientation sensors, as well as an active needle guidance mechanism, the device enables novice clinicians to build the skills and knowledge necessary to safely perform needle-based procedures. This device has the potential to accelerate the training curve, improve skill retention, and reduce biopsy complication rates in clinical settings.

B. Innovation. This project introduces a novel approach to biopsy training that combines real-time feedback with physical correction in a compact, low-cost platform with features not currently found in standard training tools[6]. The integration of an inertial measurement unit (IMU) and pressure sensors allows for immediate detection of improper needle angle and excessive insertion force[7]. The device then communicates errors through visual (LEDs) cues, creating a multisensory learning environment.

Unlike passive simulators, this device employs a motorized system that dynamically adjusts needle orientation during insertion[8]. This active correction provides users with an understanding of proper technique and reinforces correct needle placement during early-stage learning. The system is designed using affordable microcontrollers and 3D-printed components, making it scalable for widespread deployment in resource-limited educational institutions and global training programs. By merging tactile feedback with smart guidance, this device represents a significant advancement in procedural simulation technology and a step toward safer, more efficient medical training.

C. Approach.

To develop this instrument, the works are divided into four subsystem developments: ergonomics, angle correction (actuation and sensors), and user feedback. The development of the ergonomic subsystem is based on the consideration of the user's comfort while holding, being lightweight, and being easy to use [9] [10]. The actuation subsystem uses a stepper motor with a parallel linkage to perform physical correction for the angle of insertion of the biopsy needle[11][12]. The sensory subsystem includes two IMU sensors and one force-sensing resistor (FSR). One of the IMU sensors tracks the orientation of the biopsy device and returns the angle relative to the insertion surface to the control system, which then controls the actuation subsystem to correct the angle of insertion[13]. Lastly, the user feedback subsystem uses an LCD screen to display

information derived from the second IMU which tracks the orientation of the user's hand and the FSR[14]. The feedback provided to the user is the deviation from the desired angle along with the force measured through the FSR[15]. It is expected that the angle correction from the control system will help stabilize the needle while performing insertion of the biopsy device and the user feedback will improve the performance of novice surgical technicians[16].

Aim 1: Develop a sensor-integrated biopsy training device. To accomplish Aim 1, we designed and manufactured a sensor-integrated biopsy training device. The device is constructed using a 3D-printed housing that mounts the electrical components and the actuation mechanism. The mechanism uses a parallel linkage attached to the stepper motor. The parallel linkage allows for the translation of the rotational pivot, saving space at the needle connection. The device is also embedded with two IMUs (MPU-6050) and FSR that is positioned along the needle path. The IMUs will continuously monitor the needle's orientation and the user's hand orientation relative to the surface of the operation region, while the FSRs will detect axial force exerted during needle insertion. A microcontroller (Arduino Nano) will process sensor inputs and control the actuation subsystem as well as display feedback to the user. The control system would try to stabilize the insertion angle of the needle at 52.5 degrees relative to the surface of the target region [17]. A red LED is attached to the device to warn the user when the force exceeds the threshold of 5 newtons and the error angle is beyond 7.5 degrees. These thresholds reflect clinically relevant ranges associated with suboptimal needle technique for lung and liver biopsies (5 Newtons and 45-60 degrees on insertion angle)[18].

The electrical components are tested for functionality on a breadboard before soldering and fixed in place inside the 3D-printed housing. The needle and hand orientations are derived from the instantaneous angular velocities and accelerations from the two IMUs separately using functions from the MPU6050 library on Arduino IDE. Because we only need the orientation from one axis of the IMU feedback, we only tested it in one direction of rotation. The result shows that the orientation tends to drift over time due to the accumulated noise [19]. The FSR feedback is an analog voltage feedback, which is calibrated using a known weight and linear regression to predict the force. It is also worth mentioning that the IMUs and the LCD screen communicate with the Arduino Nano through an I2C communication protocol, which saves space inside the housing[20].

Limitation and future improvement: The limitation of the orientation feedback is that it is relative to its starting position. To make use of this device, it is required that the device is placed down for sensor calibration before beginning the operation. The testing of the IMUs was rather qualitative than quantitative, and no accuracy testing was performed, so to further develop this device, more detailed testing and analysis have to be done and a possible calibration may need to be performed for each IMU sensor[21]. The FSR is also calibrated with only 2 points (0 force and 2N force), which does not capture the whole spectrum of the reading, which can be nonlinear. We only wanted a rough estimation because that was not our focus for the project, but to implement the device effectively, further calibration is needed.

Aim 2: Implement real-time visual feedback based on insertion angle and applied pressure. The LCD screen displays the deviation of the needle angle from the desired needle angle and the pressure through the needle sensed by the FSR. This step is implemented by using LiquidCrystal_I2C library with Arduino IDE and calibrated force feedback discussed in Aim 1. Ideally, the LCD screen would be mounted onto the top of the device where the user can see and adjust their hand based on the feedback as part of the learning process. However, for the demonstration purpose, the LCD screen is mounted onto the side of the device. A red LED is used to warn the user when the applied pressure is higher than the save threshold as discussed in Aim 1.

Aim 3: Implement an active guidance mechanism based on insertion angle and applied pressure and evaluate the performance of the users. To extend the functionality of the device, an active guidance system that physically corrects needle angle in real time using mechanical actuation discussed in Aim 1 is incorporated. This mechanism will consist of a stepper motor connected to an actuation system embedded within a custom 3D-printed needle holder. The IMU data will be used to stabilize the angle of insertion at 52.5 degrees with the maximum range of motion between -15° to 15° . This limited range of motion is due to the design of the housing. The microcontroller uses a proportional-integral controller (PI-controller) to stabilize the needle within the operational range[22]. The PI controller is tuned for quick response time and elimination of steady-state error. This mechanism is expected to improve the performance of the users too [23].

To evaluate the effectiveness of the active guidance feature, we will perform needle insertions into the foam under three conditions: baseline (no feedback or guidance), feedback-only (using the system developed in Aim 1), and active guidance (with the motor-driven correction mechanism enabled). Performance metrics will include average angular deviation from the desired angle, task completion time, maximum force, and NASA task load [24]. We anticipate that the active guidance condition will yield the greatest improvements in user performance, particularly among novice users, by reducing needle angle deviation and facilitating faster learning[25]. Based on the three testing conditions, the result shows that the average angular deviations are 15.34, 5.23, and 5.21 degrees, respectively; the task completion times are 3.12, 4.21, and 6.73 seconds; the maximum forces are 12.98, 5.21, and 4.67 Newtons; And the NASA task loads are 61.3, 43.7, and 38.

The results show improvement in the user's performance among the first three matrices when the visual feedback is active and when the angle correction is active. There is no clear distinction in terms of precision improvements between when the active guidance is active and inactive. However, this can be due to the flaws introduced by the controller. The task completion time for each test does not vary a lot due to the nature of the operation. The increased time when the feedback is active is likely due to the time users take to try to minimize the error and the increased time when the active guidance is active is due to the motor control delaying the feedback response. The NASA task load also increases when the two assistive features are inactive. This shows the benefits of having the feedback [26]. Overall, these aims enable the development of a novel biopsy training tool that not only simulates realistic tissue interaction but also actively teaches safe procedural techniques through real-time sensory and mechanical feedback. This innovation has the potential to significantly improve procedural training outcomes and patient safety in clinical practice[27].

Limitation and future improvement: The range of motion of the active guidance is 15 degrees, which can be improved by redesigning the housing. The range of motion is only possible in one degree of freedom, which is not applicable in real life where there are up to 6 degrees of freedom (3 for translation and 3 for rotation)[28]. However, this design is a proof-of-concept that is useful for novice physicians[29]. To extend the application of this device to not rely on a physician's physical movement at all, the device would need more degree of freedom to stabilize the needle. The PI controller also suffers from cumulative errors from the IMUs, which creates SSE over time. The desired angle is currently being manually input into the Arduino code, which makes it difficult for future changes for different applications. It is recommended that this number be adjusted through a user interface.

The evaluation varies significantly among different test subjects, which only have five people, and the test subjects are not certified physicians[30]. This can introduce bias and does not guarantee a conclusion of improvement. In future iterations, it is recommended that the test is conducted with more test subjects who also have experience with the biopsy/insertion operation.

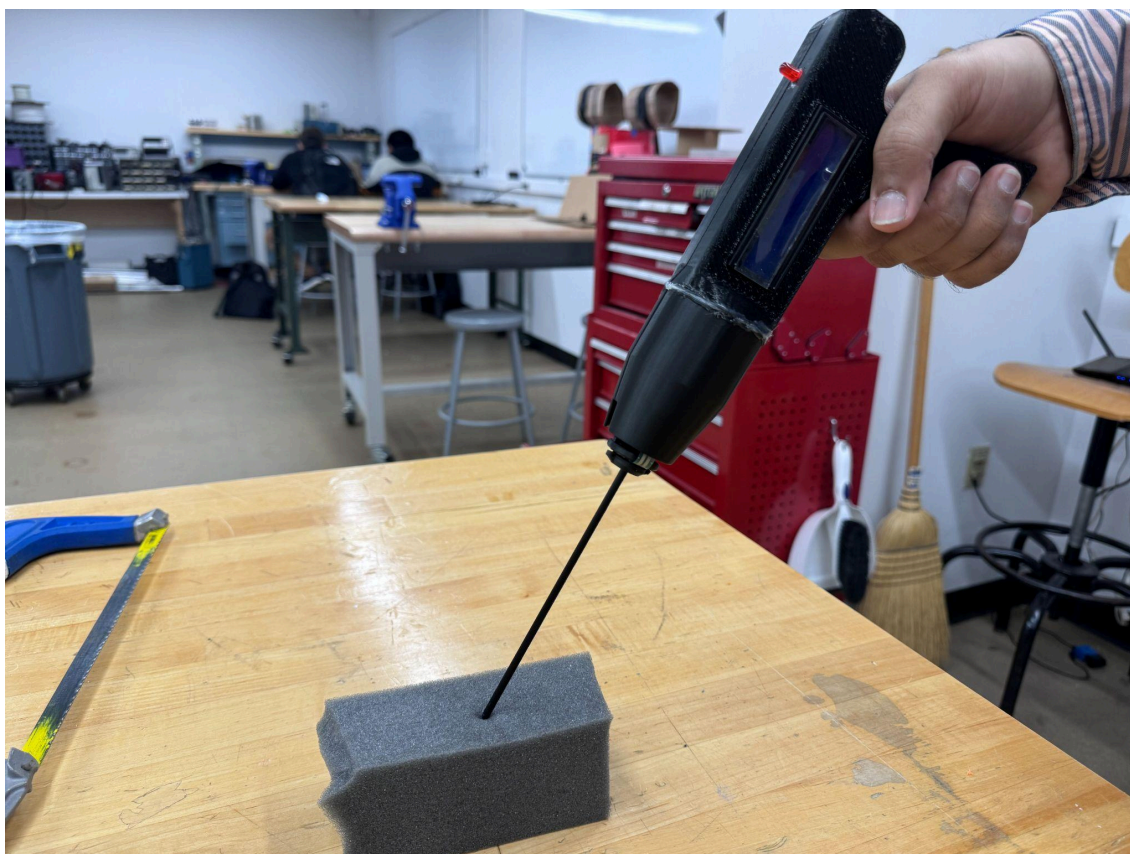


Figure 1: Demonstration of Robotic-assisted Laparoscopic Device in Testing Environment

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